

Photon deflects upwards towards plate 1.

7 (a) (i) p of electron $= \sqrt{2mE_k}$
 $= \sqrt{2(9.11 \times 10^{-31})(4.96 \times 10^{-24})}$
 $= 3.0 \times 10^{-27} \text{ kg m s}^{-1}$

(ii) de Broglie wavelength

$$\lambda = \frac{h}{p}$$
$$= \frac{6.63 \times 10^{-34}}{3.0 \times 10^{-27}}$$
$$= 2.2 \times 10^{-7} \text{ m}$$

(b) (i) An electron in an energy level is bound to the nucleus (or atom) by the attractive Coulomb force. The total energy of such a bound system is negative.

(ii) 9.0 eV electron has sufficient energy to excite mercury atom from ground state to energy level of -1.57 eV (because $-1.57 - (-10.38) = 8.81 \text{ eV}$).

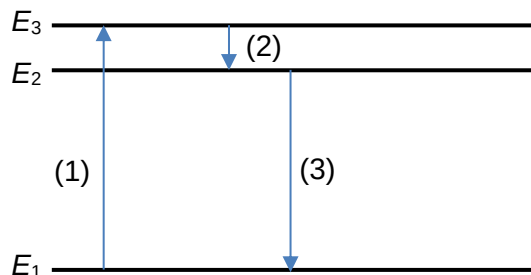
As atom de-excites from -1.57 to -5.74 eV , a photon is emitted.

$$\frac{hc}{\lambda} = [-1.57 - (-5.74)] \times (1.6 \times 10^{-19})$$

$$\lambda = \frac{(6.63 \times 10^{-34})(3.00 \times 10^8)}{6.672 \times 10^{-19}}$$

$$\lambda = 2.98 \times 10^{-7} \text{ m}$$

(iii)



8 (a) Simple harmonic motion is defined as the motion of a particle (or body) about a fixed